PERMUTATIONS & COMBINATIONS-II

1. How many ways the word DETAIL be arranged in such a way that vowels occupy only odd positions?
2. How many ways the word MACHINE be arranged in such a way that vowels occupy only odd positions?
3. How many ways 4 boys and 3 girls can be arranged in a row?

i) If both ends must be occupied by boys

ii) If all girls always sit side by side

iii) If all girls do not sit come together

iv)If no two girls sit come together

iv) If no two boys sit come together

v) If boys and girls sit alternatively

4. How many 3-digit numbers can be formed from the digits 1,2,3,4 and 5 assuming that
   (i) Repetition of the digits is allowed?
   (ii) Repetition of the digits is not allowed?
5. How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67 and no digit appears more than once?
6. How many three-digit numbers are there in which all the digits are odd?
7. How many three-digit numbers are there in which all the digits are even?
8. How many 2 digit even numbers can be formed from the digits 1,2,3,4,5 if the digits can be repeated?
9. How many 4-digit number can be formed using the digits (0,1,2,3,4)?
10. How many odd numbers less than 2000 can be formed using the digit 0,1,3,4,8,7 if repetition of digits is allowed?
11. How many numbers between 1000 and 5000 can be formed with the digits 0,1,3,5,6 if repetition is not allowed?
12. In how many ways 6 persons be seated around a circular table for a dinner?
13. 12 persons are seated at a round table. How many ways we can arrange that 4 particular persons seated together?

CODEGNAN